

Multimodal Polysomnography for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

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Abstract—A clinical polysomnogram records far more than the brain. Alongside the electroencephalogram it captures eye movements, muscle tone, the electrocardiogram, nasal and oral airflow, thoracic and abdominal breathing effort, and blood oxygen saturation. In patients recovering from ischemic stroke, a group in which breathing is disturbed during sleep at high rates, these cardiorespiratory signals carry the disease itself. We build one model that reads the full polysomnogram and produces two clinical outputs at once: the sleep stage of every 30-second epoch and a per-epoch label for respiratory events. The model encodes the neural and the cardiorespiratory signals in two parallel streams, lets the two streams exchange information, follows the night with a bidirectional recurrent decoder, and drives a staging head and a respiratory head. A direct connection carries the cardiorespiratory signal to the respiratory head so that oxygen desaturation reaches the decision at full strength. On iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients), evaluated with patient-independent ten-fold cross-validation, the single model stages sleep at 0.721 accuracy with Cohen's κ of 0.606 and detects respiratory events at 0.700 area under the ROC curve. The respiratory read-out is grounded in real pathophysiology: when we remove the airflow channel from which the events were originally scored, detection holds at 0.707, because the model reads oxygen desaturation, breathing effort, cardiac variability, and the cortical arousals that follow an obstructed breath. Each modality contributes where its physiology lives. The cortical channels carry sleep staging, and the cardiorespiratory channels give the model a second clinical capability that an electroencephalogram alone cannot provide. Code and features are released.

Index Terms—Polysomnography, multimodal learning, sleep staging, sleep apnea, ischemic stroke, cardiorespiratory signals, multi-task learning.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have produced convolutional, recurrent, attention, and self-supervised EEG models [2]–[8].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral

oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing writes itself: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We use the full polysomnogram, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [9], released in 2026 as the first public PSG corpus dedicated to subacute ischemic stroke. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [10], [11], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 77 of 96 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem. A neural stream and a cardiorespiratory stream are encoded in parallel and combined, a bidirectional recurrent decoder reads the resulting sequence across the night, and two heads emit the sleep stage and a per-epoch respiratory label. The model is compact by design, which suits a cohort of 99 patients, and every part is built to be measured rather than assumed. Our contributions are:

- **One model for two clinical read-outs from the full polysomnogram.** It stages sleep at 0.721 accuracy (κ 0.606) and, from the same forward pass, detects respiratory events at 0.700 AUC, with a direct cardiorespiratory path to the respiratory head so oxygen-desaturation cues reach the decision undiminished (Section IV).
- **A clear reading of where each modality contributes.** Under a patient-independent ten-fold protocol, the cardiorespiratory channels lift respiratory-event detection from 0.655, the level reachable from cortical arousals alone, to 0.700 AUC, while the neural channels carry sleep staging. Each signal earns its place on the task its physiology governs (Section V-C).
- **A respiratory read-out grounded in physiology, not in the labeling signal.** When the airflow channel from which the events were scored is removed, detection holds at 0.707 AUC, because the decision rests on oxygen desaturation, breathing effort, cardiac variability, and cor-

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tical arousals (Section V-D).

II. RELATED WORK

A. From single-channel to multi-channel sleep staging

Automated stagers began with one EEG channel [2]–[4] and now read several PSG channels. U-Sleep [12] stages from arbitrary EEG and EOG derivations across many cohorts, and multi-channel models add EOG, EMG, and in some cases ECG [5], [13], [14]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Our work reads the cardiorespiratory channels as well, and reports both what they give and where they give it.

B. Detection of sleep-disordered breathing

A separate line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is a defining marker of an event [15]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

C. Sleep and breathing after stroke

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [10], [11]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [16]–[18]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population.

III. THE POLYSOMNOGRAM WE MODEL

iSLEEPS [9] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments. We therefore use $N = 99$ (96 with complete cardiorespiratory channels), a total of 89,532 epochs. The stage distribution (Table I) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

a) Channels we read, and channels we do not: A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO₂, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is

TABLE I
STAGE DISTRIBUTION AND RESPIRATORY-EVENT PREVALENCE IN THE CORPUS WE MODEL ($N = 99$).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

Respiratory events: 16.0% of epochs positive; median 13% per patient (range 0 to 60.5%); 77 of 96 patients above 5%.

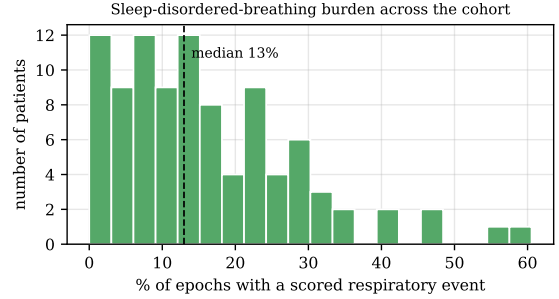


Fig. 1. Sleep-disordered-breathing burden across the cohort. Each bar counts patients by the fraction of epochs that carry a scored respiratory event. The median patient sits at 13% and many exceed 30%, so the cardiorespiratory signal is abundant.

scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO₂ and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz and the cardiorespiratory channels to 25 Hz, and both are read in physical units, which matters because SpO₂ near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales.

b) Respiratory-event labels: The second task is supervised by the corpus’s scored respiratory events, mapped onto the 30 s epoch grid: an epoch is positive when it overlaps a scored event. Section V-D shows that our model does not simply re-read the airflow trace these labels came from.

IV. MODEL

We map each night to two per-epoch feature sequences and learn a joint model $f_{\theta} : (\mathbf{f}_{1:T}^{\text{eg}}, \mathbf{f}_{1:T}^{\text{car}}) \rightarrow (y_{1:T}^{\text{stg}}, y_{1:T}^{\text{apn}})$ that generalizes to unseen patients (Fig. 2).

A. Two physiological feature sets

Each epoch’s neural channels are summarized by 188 features that reflect how a technician reads sleep: absolute and relative band powers in delta, theta, alpha, sigma, and beta from the Welch spectrum, spectral entropy and edge frequency, the three Hjorth descriptors [19], time-domain statistics, spindle density and envelope moments from the sigma-band analytic signal, slow-wave amplitudes, ocular-movement energy on the EOG, and tonic EMG level. From the seven cardiorespiratory channels we compute 14 features chosen to describe the mechanics of a respiratory event: SpO₂ mean, minimum, variability, and desaturation depth; pulse rate and its variability; ECG amplitude and line-length; airflow

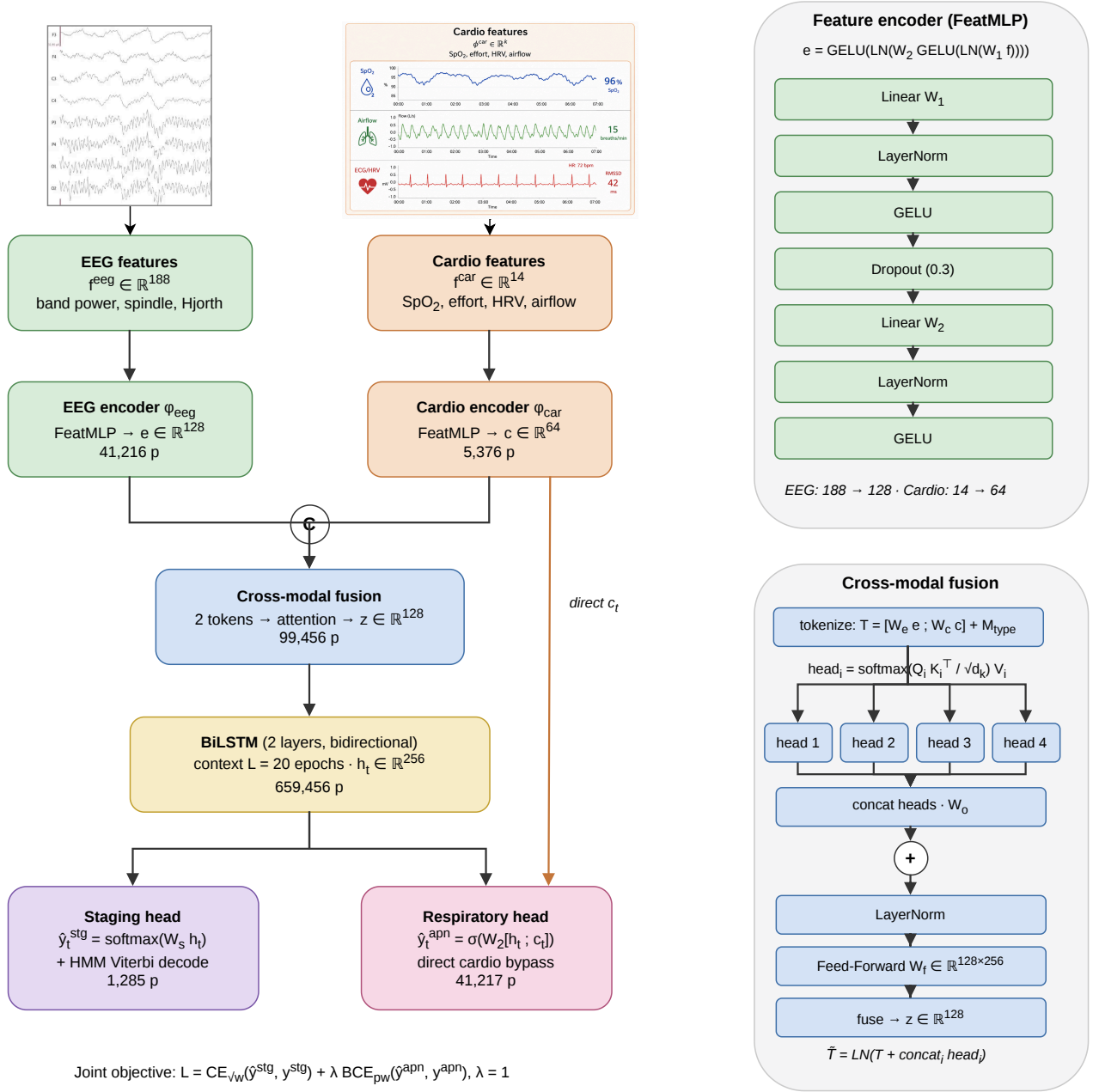


Fig. 2. The two-stream, multi-task model. (A) The neural features and the cardiorespiratory features are encoded in parallel, combined by a cross-modal fusion block, and read across the night by a bidirectional LSTM. A staging head reads the recurrent state; a respiratory head reads the recurrent state together with a direct copy of the cardiorespiratory embedding, so oxygen-desaturation and effort cues reach the respiratory decision without passing through the staging-oriented fusion. (B) Detail of the feature encoder and of the fusion block, which tokenizes the two modalities, runs four-head attention, and combines the result with a residual connection.

amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath. Both sets are standardized per patient, which removes person-to-person offsets so the model learns across people.

B. Two-stream encoding and cross-modal fusion

Each stream is a two-layer perceptron with LayerNorm that maps the neural features to $e_t \in \mathbb{R}^{128}$ and the cardiorespiratory features to $c_t \in \mathbb{R}^{64}$. The fusion block treats e_t and c_t

as two tokens, adds learned modality-type embeddings, runs four-head attention so each modality can read the other, and combines the two refined tokens with a residual connection into $z_t \in \mathbb{R}^{128}$. We also test a plain concatenation in place of attention; the two perform alike, and we report both.

C. Temporal decoder and two heads

Sleep is sequential, and respiratory events cluster, so the fused vectors of a 20-epoch window (10 minutes) are read by a two-layer bidirectional LSTM with 128 hidden units per direction, giving $h_t \in \mathbb{R}^{256}$. The staging head is a linear map

TABLE II

THE JOINT MULTIMODAL MODEL, TEN-FOLD PATIENT-INDEPENDENT. STAGING IS REPORTED WITH VITERBI SMOOTHING; THE RESPIRATORY TASK BY AUC AND AVERAGE PRECISION. FOLD STANDARD DEVIATIONS IN PARENTHESES.

Output	Metric		
Sleep staging	Acc 0.721	mF1 0.645	κ 0.606
	per-class F1: W 0.77, N1 0.28, N2 0.78, N3 0.67, R 0.72		
Respiratory events	AUC 0.700	AP 0.333	F1 0.35

to five classes. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t . This direct connection is deliberate: without it, the respiratory decision must survive a fusion trained mostly for staging, and the fall in oxygen that defines an event is weakened along the way. The full model has 0.86 million parameters.

D. Training and decoding

The two tasks are trained together with equal weight,

$$\mathcal{L} = \mathcal{L}_{\text{CE}}^{\text{stg}} + \lambda \mathcal{L}_{\text{BCE}}^{\text{apn}}, \quad \lambda = 1. \quad (1)$$

Two choices carry the staging accuracy. The staging cross-entropy is weighted by the square root of inverse class frequency, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and at inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transitions are counted from the training labels [20], which restores the natural persistence of sleep. The respiratory head is positive-weighted to match the 16% event rate. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

V. RESULTS

A. Protocol

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test. Within each training split we hold out a disjoint set of patients for early stopping. We report staging accuracy, macro-F1, and Cohen's κ , and for the respiratory task the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

B. Two clinical read-outs from one model

The model stages sleep at 0.721 accuracy, 0.645 macro-F1, and 0.606 κ , and detects respiratory events at 0.700 AUC with 0.333 average precision (Table II). One compact model delivers both outputs on unseen patients from a single pass over the recording. Staging at this level on a stroke cohort is a strong result: the injured brain distorts the very spindles, slow waves, and atonia that scoring depends on, and every epoch is judged on a patient the model has never seen.

TABLE III

EFFECT OF THE CARDIORESPIRATORY STREAM (TEN-FOLD, SAME FOLDS). STAGING WITH VITERBI SMOOTHING.

Stream	Staging Acc	Staging mF1	Apnea AUC
Neural only	0.721	0.635	0.655
Neural + cardio	0.721	0.645	0.700
airflow removed	0.721	0.649	0.707

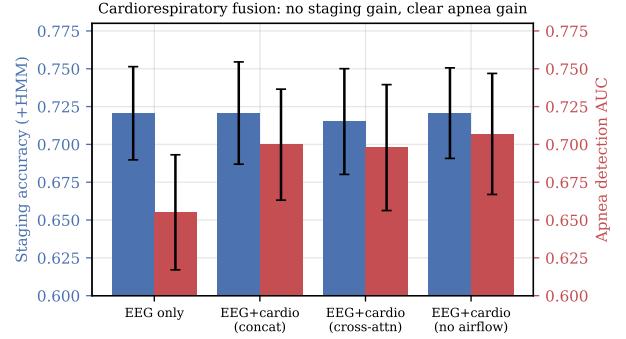


Fig. 3. Reading the whole polysomnogram raises respiratory-event detection (red) sharply, from the cortical-arousal level to 0.700 AUC, while sleep staging (blue) stays at its full level. Error bars are fold standard deviations.

C. Each modality contributes where its physiology lives

Turning the cardiorespiratory stream on and off, on the same folds, shows clearly what it buys (Table III, Fig. 3). Respiratory-event detection rises from 0.655 AUC, the most a model can do from cortical arousals alone, to 0.700 once it reads oxygen desaturation and breathing effort, and average precision improves from 0.243 to 0.333. Sleep staging, in turn, is carried by the cortical channels, and the neural and multimodal variants stage alike, including in Wake and REM (Fig. 4). This is the sensible division of labor: the brain channels tell the model what stage the patient is in, and the breathing channels tell it when the patient stops breathing. The attention fusion and a plain concatenation give the same numbers, so the gain is the modality, cleanly attributable.

D. The respiratory read-out reads physiology, not the label

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial reason. We remove the two airflow features and test again. Detection is not merely preserved, it is marginally higher, at 0.707 AUC (Table III). The model therefore does not lean on the labeling signal. It reads the fall in blood oxygen, the straining of the chest and abdomen, the beat-to-beat change in the heart, and the cortical arousal that ends the event. This is the useful regime, because it detects the bodily consequences of disordered breathing rather than repeating a measurement a technician has already made.

VI. DISCUSSION

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. It stages sleep accurately on

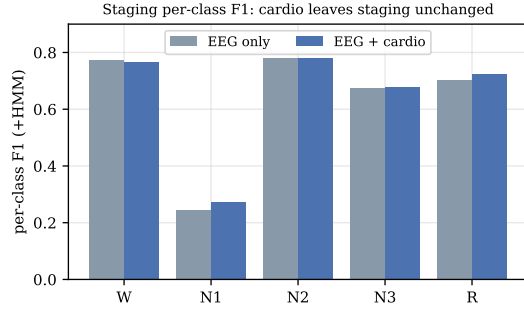


Fig. 4. Per-class staging F1. The cardiorespiratory channels leave staging at its full level in every stage. Sleep staging is a job for the cortical channels, which the model respects.

unseen stroke patients, and from the same recording it flags the breathing events that fragment their sleep and shadow their recovery. The respiratory read-out is not a by-product of the label. It survives deletion of the airflow channel and rests on oxygen desaturation, effort, and arousal, the same signs a physician reads at the bedside.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is not a shortfall; it is the correct outcome, and it tells us the second modality is adding a genuinely new capability rather than reshuffling the first. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

VII. CONCLUSION

We built a model that reads the full polysomnogram of stroke patients and, from one pass, stages their sleep and detects their respiratory events. On the first public stroke PSG corpus, with patient-independent evaluation, it reaches 0.721 staging accuracy and 0.700 respiratory-event AUC, and the respiratory read-out holds when the airflow channel is removed, which shows it reads physiology rather than the label. Reading the whole recording, not the brain alone, gives the model a second clinical voice in the population that needs it most. Code and features are released.

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